

$$1) a) \sum_{k=0}^{\infty} 5 \cdot \left(\frac{1}{2}\right)^k = 5 \cdot \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = 5 \cdot \frac{1}{1 - \frac{1}{2}} = 5 \cdot 2 = 10$$

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

$$b) \sum_{k=2}^{\infty} 2 \cdot (-0,4)^k = 2 \cdot \sum_{k=2}^{\infty} \left(-\frac{2}{5}\right)^k = 2 \cdot \sum_{k=0}^{\infty} \left(-\frac{2}{5}\right)^{k+2} = 2 \cdot \sum_{k=0}^{\infty} \left(-\frac{2}{5}\right)^2 \cdot \left(-\frac{2}{5}\right)^k$$

oder

$$= 2 \cdot \frac{4}{25} \cdot \sum_{k=0}^{\infty} \left(-\frac{2}{5}\right)^k = \frac{8}{25} \cdot \frac{1}{1 - (-\frac{2}{5})} = \frac{8}{25} \cdot \frac{5}{7} = \frac{8}{35}$$

$$\rightarrow 2 \cdot \left( \sum_{k=0}^{\infty} \left(-\frac{2}{5}\right)^k - \left( \left(-\frac{2}{5}\right)^0 + \left(-\frac{2}{5}\right)^1 \right) \right) = 2 \cdot \left( \frac{1}{1 - (-\frac{2}{5})} - \frac{3}{5} \right) = 2 \cdot \left( \frac{5}{7} - \frac{3}{5} \right) = 2 \cdot \frac{25-21}{35}$$

$$2) a) \sum_{k=1}^{\infty} \sqrt{k} \rightarrow \text{divergent, da } \lim_{k \rightarrow \infty} \sqrt{k} \neq 0 \text{ (Keine Nullfolge)}$$

$$b) \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \rightarrow \text{divergent, da } \frac{1}{\sqrt{k}} \geq \frac{1}{k} \text{ und } \sum_{k=1}^{\infty} \frac{1}{k} \text{ divergent ist (harmonische Reihe)}$$

$$c) \sum_{k=1}^{\infty} (-1)^{k+1} \cdot \frac{1}{\sqrt{k}} \quad \lim_{k \rightarrow \infty} \frac{1}{\sqrt{k}} = 0 \Rightarrow \text{konvergent} \quad \text{Leibniz } \sum_{k=1}^{\infty} (-1)^{k+1} \cdot a_k, \quad \lim_{k \rightarrow \infty} a_k = 0$$

$$d) \sum_{k=1}^{\infty} \left(\frac{1}{\sqrt{k}}\right)^k \quad \lim_{k \rightarrow \infty} \sqrt[k]{\left(\frac{1}{\sqrt{k}}\right)^k} = 0 < 1 \Rightarrow \text{konvergent} \quad \text{Wurdsatz: } \lim_{k \rightarrow \infty} \sqrt[k]{a_k} < 1$$

3) a)  $\sum_{k=2}^{\infty} \frac{k}{3^k} \rightarrow$  Wurzelsatz:  $\lim_{k \rightarrow \infty} \sqrt[k]{\frac{k}{3^k}} = \frac{\sqrt[k]{k}}{3} = \frac{1}{3} < 1 \Rightarrow$  konvergent

Quotientensatz:  $\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1 : \lim_{k \rightarrow \infty} \frac{(k+1) \cdot 3^k}{3^{k+1} \cdot k} = \left(1 + \frac{1}{k}\right) \cdot \frac{1}{3} = \frac{1}{3} < 1$

b)  $\sum_{k=2}^{\infty} (-1)^k \cdot (1 - \frac{1}{k})$ , da  $\lim_{k \rightarrow \infty} (1 - \frac{1}{k}) = 1 \neq 0$  divergent (Leibniz)

c)  $\sum_{k=1}^{\infty} \frac{2^{k+2}}{k^2 \cdot 5^k} \rightarrow$  Wurzelsatz:  $\lim_{k \rightarrow \infty} \sqrt[k]{\frac{2^{k+2}}{k^2 \cdot 5^k}} = \frac{2 \cdot 1}{1 \cdot 5} = \frac{2}{5} < 1 \Rightarrow$  konvergent

Quotient:  $\lim_{k \rightarrow \infty} \frac{2^{(k+1)+2} \cdot 5^k \cdot k^2}{(k+1)^2 \cdot 5^{k+1} \cdot 2^{k+2}} = \frac{2^{k+3}}{2^{k+2}} \cdot \frac{5^k}{5^{k+1}} \cdot \frac{k^2}{(k+1)^2} = \frac{2}{5} \cdot \left(\frac{k}{k+1}\right)^2 = \frac{2}{5} < 1$

d)  $\sum_{k=1}^n \left( \frac{1}{\sqrt{k+1}} - \frac{1}{\sqrt{k}} \right) = \frac{1}{\sqrt{n+1}} - 1 \rightarrow \lim_{n \rightarrow \infty} \left( \frac{1}{\sqrt{n+1}} - 1 \right) = -1 \Rightarrow$  konvergent

| k   | $\frac{1}{\sqrt{k+1}}$ | $\frac{1}{\sqrt{k}}$   | $\frac{1}{\sqrt{k+1}} - \frac{1}{\sqrt{k}}$ |
|-----|------------------------|------------------------|---------------------------------------------|
| 1   | $\frac{1}{\sqrt{2}}$   | 1                      | $\frac{1}{\sqrt{2}} - 1$                    |
| 2   | $\frac{1}{\sqrt{3}}$   | $\frac{1}{\sqrt{2}}$   | $\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2}}$   |
| 3   | $\frac{1}{\sqrt{4}}$   | $\frac{1}{\sqrt{3}}$   | $\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{3}}$   |
| ... | ...                    | ...                    | ...                                         |
| n-1 | $\frac{1}{\sqrt{n}}$   | $\frac{1}{\sqrt{n-1}}$ | $\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n-1}}$ |
| n   | $\frac{1}{\sqrt{n+1}}$ | $\frac{1}{\sqrt{n}}$   | $\frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n}}$ |

}  $\frac{1}{\sqrt{n+1}} - 1$

$$4) \quad x_{n+1} = \sqrt{2 + x_n} \quad ; \quad x_0 = 1$$

Monotonie Tendenz:  $x_1 = \sqrt{2+1} = \sqrt{3} > 1 = x_0$  Behauptung:  $x_{n+1} > x_n$

$$n=0 : x_1 > x_0 : \sqrt{3} > 1 \quad \checkmark$$

$$n := n+1 : x_{n+2} > x_{n+1} \quad \text{zu zeigen}$$

$$x_{n+1} > x_n \quad | +2$$

$$x_{n+1} + 2 > x_n + 2 \quad | \sqrt{\phantom{x}}$$

$$\underbrace{\sqrt{x_{n+1} + 2}}_{x_{n+2}} > \underbrace{\sqrt{x_n + 2}}_{x_{n+1}}$$

$\Rightarrow$  Die Folge ist streng monoton steigend und somit  $x_0=1$  untere Schranke.

Schranken: Behauptung:  $x_n \leq 2$

$$n:=0 : x_0 = 1 \leq 2 \quad \checkmark$$

$$n := n+1 : x_{n+1} \leq 2 \quad \text{zu zeigen}$$

$$x_n \leq 2 \quad | +2$$

$$x_n + 2 \leq 4 \quad | \sqrt{\phantom{x}}$$

$$\underbrace{\sqrt{x_n + 2}}_{x_{n+1}} \leq \sqrt{4} = 2$$

$$x_{n+1} \leq 2$$

$\Rightarrow$  Die Folge ist beschränkt  $x_n \in [1, 2]$  und somit konvergent und der Grenzwert existiert. SHOW

$$\lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} x_n = \gamma$$

$$\Rightarrow \gamma = \sqrt{2 + \gamma} \quad | \uparrow^2$$

$$\gamma^2 = 2 + \gamma \quad | -2 - \gamma \quad \rightarrow \text{Satz von Viete}$$

$$\gamma^2 - \gamma - 2 = (\gamma - 2)(\gamma + 1) = 0$$

$$\gamma = 2 \vee \gamma = -1$$

$$\Rightarrow \lim_{n \rightarrow \infty} x_{n+1} = 2$$

5) a)  $(-x^3 + 4x^2 - x - 6) : (x - 2) = -x^2 + 2x + 3 = -(x^2 - 2x - 3) = -(x - 3)(x + 1)$

$$\begin{array}{r} -(-x^3 + 2x^2) \\ \hline 2x^2 - x - 6 \\ -(2x^2 - 4x) \\ \hline 3x - 6 \\ -(3x - 6) \\ \hline \end{array}$$

$\rightarrow -(x - 2)(x - 3)(x + 1) = 0 \Rightarrow \mathbb{L} = \{-1; 2; 3\}$

b)  $(3x^3 + 10x^2 - 7x + 4) : (3x^2 - 2x + 1) = x + 4$

$$\begin{array}{r} -(3x^3 - 2x^2 + 1) \\ \hline 12x^2 - 8x + 4 \\ -(12x^2 - 8x + 4) \\ \hline \end{array}$$

c)  $(x^5 - 2x^3 - x^2 + 1) : (2x^3 - 2) = \frac{1}{2}x^2 - 1 + \frac{-1}{2x^3 - 2}$

$$\begin{array}{r} -(x^5 - x^2) \\ \hline -2x^3 + 1 \\ -(-2x^3 + 2) \\ \hline -1 \end{array}$$